

Topology

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Let us assume that Space – the set of all points – exists, and satisfies the axioms stated below. Denote Space by S .

Axiom 1: There exist a collection of subsets of S called *regions* such that (1) every point lies in some region (the regions “cover” S), and (2) if the point p lies in each of the regions R_1 and R_2 then there is a region R_3 such that $p \in R_3 \subset (R_1 \cap R_2)$.

Definition: The point q is a *limit point* of the point set M provided that every region containing q contains a point of M other than q . The point set a is *closed* provided that it contains all of its limit points. The set B is *open* iff $S - B$ is closed.

Proposition 1: Every region is open.

Proposition 2: A (non-empty) set is open iff it is a union of regions.

Proposition 3: The union or intersection of 2 closed sets is closed.

Proposition 4: The union or intersection of 2 open sets is open.

Proposition 5: Every finite set is closed.

Question 1: How can Propositions 3 and 4 be generalized?

Axiom 2: For any two points p and q , there are disjoint open sets U and V such that $p \in U$ and $q \in V$.

Axiom 3: For any point p and any closed point set M not containing p , there are disjoint open sets Q and R containing p and M respectively.

Lemma: Axiom 3 implies that if p is in an open set R , then there exists an open set Q such that $p \in Q \subset \overline{Q} \subset R$.

Question 2: Do Axioms 1 and 3 imply Axiom 2?

Definition: If M is a point set, then M' (the derived set of M) is the set of all limit points of M , and $\overline{M} = M \cup M'$. ($\overline{M}$ called the *closer* of M .)

Lemma: If p is a limit point of $H \cup K$, then p is a limit point of at least one of H and K .

Proposition 6: If M is a point set, then $\overline{M}$ is closed.

Axiom 4: There is a distance function d for S such that (1) for each x and y in S , $d(x, y) = d(y, x) \geq 0$; and (2) for any point x and set M , $\inf\{d(x, y) | y \in M\} \equiv 0$ iff $x \in \overline{M}$.

Proposition 7: Part (2) of Axiom 4 is equivalent to: (1) for any x and y in S , $d(x, y) = 0$ iff $x = y$, together with (2) for any point p and positive number c , there is a region R such that $p \in R \subset \{x | d(x, p) < c\}$; and conversely, for any region R containing the point p , there is a positive number c such that $p \in \{x | d(x, p) < c\} \subset R$.

Notation: The set $\{x | d(x, p) < c\}$, sometimes called the c neighborhood of p (with respect to d), will be denoted by $U_c(p)$.

Definition: A space T is *compact* iff every infinite subset of it has a limit point. The space is *bicompact* iff every open cover of it has a finite subcover. A subset M of T is bicompact iff every cover of M by a collection of open sets has a finite subcollection that also covers M .

Proposition 8: Every bicompact subset of S is closed.

Proposition 9: S is bicompact iff it is compact.

Proposition 10: Suppose that S is bicompact. If for each natural number n , M_n is a closed set and $M_1 \cap M_2 \cap \cdots \cap M_n \neq \emptyset$, then $\bigcap_{n=1}^{\infty} M_n \neq \emptyset$.

Definition: Let M be a point set. The point q is a *boundary point* of M provided that every region containing q contains both a point of M and a point of $S - M$; p is an *interior point* of M if some region containing p is contained in M ; and p is an *exterior point* of M if it is neither a boundary point nor an interior point.

Proposition 11: Suppose that S is the union of countably many closed sets $M_1, M_2, \dots$. Then for some natural number n , M_n contains an interior point. (True if S is bicompact – Baire Category).

Definition: A set is *countable* provided that it can be put into one-to-one correspondence with some subset of the natural numbers; otherwise it is *uncountable*.

Definition: A point q is a *sequential limit point* of the sequence $p_1, p_2, p_3, \dots$ provided that every region containing q contains all but (at most) finitely many terms of the sequence (i.e. contains p_n for all but at most finitely many values of n).

Proposition 12: A sequence of points of S can have at most one sequential limit point (hence if q is the sequential limit point of $p_1, p_2, \dots$, we can write $\lim p_n = q$).

Example 1: The points are the real numbers and regions are all rays extending to the right (sets of the form (a, ∞) for all real numbers a).

Example 2: Points are the real numbers, and regions are all “left-handed sects” (sects of the form $[a, b)$).

Example 3: Points are real numbers, and for each real number a , $\{a\}$ is a region (*discrete* topology). (A space in which the whole space and $\emptyset$ are the only regions is called an *indiscrete* space).

Example 4: Points are the real numbers, and for each real number x and each $c > 0$, the set $R(x, c)$ consisting of x together with all rational numbers in $(x - c, x + c)$ is a region.

Example 5: Points of the space T are real numbers and R is a region iff $T - R$ is either finite or empty (finite complement topology).

Example 6: R^n , with usual metric distance function.

Example 7: Points – all sequences $x_1, x_2, x_3, \dots$ such that $|x_n| \leq 1$. Topology is that “generated” by the distance function $d|x = x_1, x_2, \dots$ and $y = y_1, y_2, \dots$ such that $d(x, y) = \sum_{n=1}^{\infty} \frac{x_n - y_n}{2^n}$.

Example 8: Let Ω denote the first uncountable ordinal, and $[1, \Omega]$ denote the set of all ordinals up to and including Ω (similarly $[1, \Omega)$ denotes those up to but *not* including Ω). For every pair of ordinals a and b , $a < b$, in $[1, \Omega]$, the segment $(a, b) = \{x | a < x < b\}$ is a region. Also, the sets $[1, a)$ and $(a, \Omega]$ are regions.

Problem 18: Show that if G is a collection of segments which covers $[1, \Omega)$, then there exists $x < \Omega$ such that $\bigcup G(x) = \bigcup \{g \in G | x \in g\}$ covers $[1, \Omega)$.

Problem 19: Determine which of Axioms 1 - 4 are satisfied by $[1, \Omega]$.

Problem 20: Determine which of Axioms 1 - 4 are satisfied by $[1, \Omega)$.

Problem 21: Show that $[1, \Omega]$ is bicomact.

Problem 22: Show that $[1, \Omega)$ is compact but not bicomact.

Proposition 23: If every uncountable subset of S has a limit point, then every uncountable subset contains a limit point.

Definition: A space T is said to be Lindelöf provided that for every open covering G of T , there is a countable subcover.

Proposition 24: If S is $\aleph_1$ -compact, then S is Lindelöf.

Definition: A space T is separable provided that there is a countable subset M of T such the $\overline{M} = T$.

Proposition 25: If S is $\aleph_1$ -compact, then S is separable.

Definition: If M is a subset of the space T , then the relative topology on M is that in which $K \subset M$ is open iff, for some open set L in T , $K = M \cap L$. In this case, we say that K “is open in M ” or “is open (rel. M)”.

Definition: The space T is hereditarily separable provided that every subset of T is separable (in its relative topology.)

Proposition 26: If S is $\aleph_1$ -compact, then S is hereditarily separable.

Proposition 27: If S is $\aleph_1$ -compact (every uncountable subset has a limit point), then S has a countable basis.

Definition: A collection B of open subsets of a space T is a *basis* for T provided that for every $p \in T$ and every open set R containing p , there is a member Q of B such that $p \in Q \subset R$. A collection C of open subsets of T is a *subbasis* for T provided that the set of all finite intersections of member of C is a basis for T .

Propositions 28, 29, 30, and 31: The converse of Prop. 24 - 27, respectively.

Example 9: Let S be the space whose points are the points of E^2 , and let a distance function d for S be defined as follows: for x and y points of S such that $\text{Im } x \cdot \text{Im } y \neq 0$, $d(x, y) = |x - y|$; and for x and y points of S such that $\text{Im } x = 0$, $d(x, y) = |x - y| + \alpha(x, y)$, where $\alpha(x, y)$ is the radian measure of the smallest non-negative angle formed by the x -axis and the line xy . Let the set of all neighborhoods $U_c(p) = \{x | d(x, p) < c\}$ be the regions for S .

Proposition 32: Suppose that the space T satisfies Axioms 1 - 3. Then T satisfies Axiom 4 iff there exists a function g from $N \times T$ into 2^T such that (1) for each x in T , $\{g_n(x) | n \in N\}$ is a local base for the topology at x (i.e., any open set containing x contains a $g_n(x)$ which contains x for some n); and (2) for x in T and any point sequence $\langle y_n \rangle$ in T such that for each n , $x \in g_n(y_n)$, the sequence $\langle y_n \rangle$ converges to x (has x as a sequential limit point).

Proposition 33: If N is a countably infinite set, then there is a collection C of subsets of N such that (1) C is uncountable; and (2) if A and B are two members of C , then A and B have at most finitely many points in common.

Proposition 34: The collection C above can be assumed to be maximal with respect to property (2) (i.e. it can be assumed that the collection C has the additional property that any subset of N which does not belong to C intersects some member of C in infinitely many elements).

Example 10: Let the points of the space T consist of the real numbers $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$, and the members of a collection C consist of subsets of $\{1, \frac{1}{2}, \frac{1}{3}, \dots\}$ satisfying Proposition 34. Regions for T will be either single points $\{\frac{1}{n}\}$ from $\{1, \frac{1}{2}, \frac{1}{3}, \dots\}$, or for each $M \in C$, the set consisting of $\{M\} \cup \{\frac{1}{m} \in M | m > n\}$ for some n .

Proposition 34.5: Example 10 satisfies Axiom 4.

Definition: A subset M of a space T is *conditionally compact* if every infinite subset of M has a limit point (in T).

Proposition 35: If M is a conditionally compact set in S , then $\overline{M}$ is compact (as a space).

Proposition 35.5: Add S normal to above hypothesis.

Definition: A space T is *normal* iff for every pair of disjoint closed sets M and N in T , there exist disjoint open sets R and Q containing M and N respectively. (A normal T_1 -space is called T_4 .)

Proposition 36: If S is bicomact, then S is normal. (Bicomact $T_2 \Rightarrow T_3 \Rightarrow T_4$)

Proposition 36.4: If T is a regular space, M a bicomact subset of T , and N a closed subset, then there exist disjoint open sets containing them.

Definition: A topological space T is *completely normal* provided that for any two mutually separated sets M and K in T , there exist mutually exclusive open sets Q and R containing M and K respectively.

Proposition 36.5: A topological space X is completely normal iff X is hereditarily normal.

Proposition 36.6: A bicomact T_2 -space is completely normal.

Proposition 36.7: A perfectly normal topological space is completely normal.

Definition: A space T is perfectly normal iff T is normal and every closed set in T is G_δ .

Proposition 36.8: Suppose that the topological space X is normal. If M and K are disjoint closed sets in X , then there exist open sets Q and R , containing M and K respectively, such that $\overline{Q} \cap \overline{R} = \emptyset$.

Definition: A function $f : S \rightarrow T$ (from a space S into a space T) is *continuous at the point x* of S provided that for every open set Q that contains $f(x)$, there is an open set R containing x such that Q contains $f(R)$. The function f is *continuous* (on S) if it is continuous at every point of S .

Proposition 37: Show that the above definition is compatible with the epsilon-delta ($\epsilon - \delta$) definition in the case of metric spaces.

Proposition 38: The function $f : S \rightarrow T$ is continuous iff for every open set Q in T , $f^{-1}(Q)$ is open in S .

Proposition 39: The continuous image of a bicomact set is bicomact.

Proposition 40: The continuous image of a Lindelöf T_2 -space is a Lindelöf T_2 -space.

Proposition 36.9: A Lindelöf T_2 -space T is hereditarily Lindelöf iff it is perfectly normal.

Proposition: A T_3 Lindelöf space is normal.

Example 11: Let S_O denote the space whose points are the points of E^2 with a basis consisting of regions of the following 3 types: (i) for each $z \in E^2$ such that $\text{Im } z \neq 0$, every open disc $\{x | |x - z|, c\}$ centered on z is a region; (ii) for each $z \in E^2$ such that $\text{Im } z = 0$ and $\text{Re } z$ is rational, each open disc counted on z is a region; and (iii) for each z such that $\text{Im } z = 0$ and $\text{Re } z$ is irrational, each “bow-tie” region (as defined in example 9) centered on z is a region.

Proposition 41: Example 11 satisfies Axioms 1 - 4, but if d is any semi-metric for S_O which is compatible with the topology of S_O , then there is some point $p \in S_O$ and some $c > 0$ such that $\{x \in S_O | d(x, p) < c\}$ is not an open set.

Example 12: The points are the points of E^1 , and a basis consists of: (i) for each irrational $p \in E^1$, $\{p\}$ is a region; and (ii) every segment (a, b) is a region (“ordinary topology” at each rational, and “discrete topology” at each irrational).

Problem 42: What can you say about Example 12?

Example 13: The points with (i) $\{p\}$ a region for each rational p ; and (ii) every segment a region (Example 12 with roles of rationals and irrationals reversed).

Problem 43: What can you say about Example 13?

Proposition 44: $[1, \Omega)$ (see Example 8) is normal.

Definition: Let A be a set. The collection C of ordered pairs each of whose terms belongs to A is a (total) *ordering* of A iff (i) C is transitive [$(a, b) \in C$ and $(b, c) \in C \Rightarrow (a, c) \in C$]; and (ii) if each of a and b belongs to A , then exactly one of $a = b$, $(a, b) \in C$ and $(b, a) \in C$ is true (“trichotomy law”). The notation “ $a < b$ ” denotes that $(a, b) \in C$. Where there is no confusion with the pair “ (a, b) ”, the segment $(a, b) = \{x \in A | a < x < b\}$ and the closed interval $[a, b] = \{x \in A | a \leq x \leq b\}$ ($[a, b)$ similarly etc.) The order topology on A generated by C is the topology for which the set of all segments is a basis, and A with this topology is an ordered space (or a *linearly ordered space*). A *partial* ordering satisfies all of the above except that a and b in A and $a \neq b$ does not necessarily imply that (a, b) or (b, a) belongs to C .

Example 14: The points of the space L are the points of the unit square $[0, 1] \times [0, 1]$, and the topology of L is the order topology induced by the lexicographical ordering (x, y precedes u, v iff either (i) $x < u$, or (ii) $x = u$ and $y < v$).

Proposition 45: The space L of Example 14 is bicomact and first countable, but not semi-metric (does not satisfy Axiom 4). Also, while every uncountable subset of L has a limit point, there is an uncountable subset of L which contains no limit point.

Definition: Let T be a topological space (space satisfying Axiom 1), and let M be a subset of T . The *relative topology* on M is the topology with respect to which a point p of M is a limit point of a subset H of M iff p is a limit point of H in T . A subset H of M is *relatively open in M* – or “*open (rel. M)*” – iff H contains no limit point of $M - H$.

Proposition 46: Let M be a subset of a topological space T . The subset H of M is relatively open in M iff there is an open subset K of T such that $H = M \cap K$.

Problem 47: Give an example of a linearly ordered space T and a subset M of T such that the relative topology on M is not the order topology generated by the induced ordering on M .

Definition: A set M is *connected* provided that there do *not* exist (non-empty) subsets H and K of M such that $M = H \cup K$ and $H \cap \overline{K} = \overline{H} \cap K = \emptyset$. (Such sets H and K with $\overline{H} \cap K = H \cap \overline{K} = \emptyset$ are called *mutually separated* sets.

Proposition 48: The set M is not connected iff M is the union of two disjoint (relatively) open and closed subsets.

Proposition 49: If M is connected, then so is $\overline{M}$.

Proposition 50: If each of M and K is connected, then so is each of $M \cup K$ and $M \cap K$.

Definition: A *continuum* is a closed and connected set (in other places, you may see the added restriction of bicomcompactness).

Proposition 51: If A is a connected set and G is a collection of connected sets each of which intersects A , then $A \cup G^*$ is connected.

Proposition 52: Every nondegenerate connected subset of S is uncountable.

Proposition 52.1: If C is a monotone collection of bicomcompact continua in a T_2 -space X , then $\bigcap C$ is a bicomcompact continuum.

Proposition 53.3: If D is a connected subset of a connected set M and $M - D = A \cup B$ where A and B are mutually separated, then $A \cup D$ is connected.

Definition: A set A is said to *separate* the connected set B if $B - A$ is not connected. The point p is called a *separating point* of the connected set M if $M - \{p\}$ is connected.

Proposition 53: Every nondegenerate compact continuum contains two non-separating points.

Example 15: Let S be the space whose points are the points of the closed upper half plane and for which R is a region, provided that R is either (i) an open disc contained in the open upper half plane, or (ii) an open disc tangent to the X -axis together with the point of tangency.

Example 16: Let T be the space whose points are the points of the closed upper half plane, for which a distance function d is defined by $d(x, y) = |x - y|$ if $(\operatorname{Im} y) > 0$ and $d(x, y) = |x - y| - \beta(x, y)$ if $\operatorname{Im} x \cdot \operatorname{Im} y = 0$ and $\beta(x, y)$ is the radian measure of the smallest nonnegative angle formed by the line xy and the vertical line through x , and let regions for T be d -neighborhoods.

Proposition 54: The spaces T and S of Examples 16 and 15 respectively are homeomorphic (i.e. there is a 1-to-1 continuous $f : S \rightarrow T$ such that f^{-1} is also continuous).

Proposition 55: The space T satisfying Axioms 1 - 3 satisfies Axiom 4 iff there is a function g from $N \times T$ into the set of all open subsets of T such that (1) for each x in T , $\{g(n, x) | n \in N\}$ is a basis for the topology at x ; and (2) if $x_1, x_2, x_3, \dots$ is a point sequence in S such that $\langle x_n \rangle \rightarrow y$, then for each m there is an n such that $y \in g(n, x_n)$.

Proposition 56: The Cartesian product of countably many metric spaces x_i ($i = 1, 2, 3, \dots$) is metric.

Definition: If for each $a \in A$, X_a is a topological space, then $\prod\{X_a | a \in A\}$, the *Cartesian product* of the spaces X_a , is the space whose points are the set of all functions $\{f : A \rightarrow \bigcup_{a \in A} X_a : \text{for each } a, f(a) \in X_a\}$, and for which a sub-basis is $\{\pi_a^{-1}(B) : B \in \mathcal{B}_a : a \in A\}$ where π_b is the projection function $\pi_b : \prod_{a \in A} X_a \rightarrow X_b$, $\pi_b(f) = f(b)$, and $\mathcal{B}_b$ is a basis for X_b for each b .